A Δ_2-free sparse-transfer theorem for the three-function Gagliardo–Nirenberg inequality

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Status. Substantial partial result, likely valid, pending human review. The result removes Δ_2 in the source's full three-N-function regime for Lebesgue measure and for an explicit sparse-transfer class of weights. The arbitrary nondoubling Hardy-weighted case remains open.

1 The source question

Kałamajska and Pietruska-Pałuba [1] consider an N-function M and functions P, Q satisfying

$$\frac{M(s)}{s^2} vw \leq M(s) + P(v) + Q(w) \qquad (s, v, w > 0). \tag{1}$$

Under a Hardy inequality for $d\mu = e^{-\varphi} dx$, they prove a modular estimate

$$\int_\Omega M(|\nabla u|)\, d\mu \leq L \int_\Omega P(\theta |\nabla^2 u|)\, d\mu + \int_\Omega Q(B|u|/\theta)\, d\mu \tag{2}$$

for every $\theta > 0$, and, when P, Q are N-functions, its multiplicative Luxemburg-norm consequence. They ask whether the hypothesis $M \in \Delta_2$ can be removed.

> **Remark 3.3** It can be deduced from the proof of Theorem 3.2 that constants L, B in the inequality (3.9) are of the form: $L = K_1 + 1$, $B = 2(\alpha_n + A + A\bar{c}(\frac{1}{A})K_2)$, where A, K_1, K_2 are the same as in (3.8), $\bar{c}$ is the same as in (2.4), α_n appears in (3.5) is such that $\alpha_n \leq c\sqrt{n}$, with $c > 0$ independent of n.
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> **Remark 3.4 (open question)** Condition $M \in \Delta_2$ in the assumptions of Theorem 3.1 is only needed to derive the formula (3.5). We do not know whether one can extend (3.2), (3.3) to functions M for which the Δ_2-conditions is not satisfied. Some results concerning the Gagliardo-Nirenberg inequalities (3.2), (3.3) hold true without the Δ_2-condition imposed on M, but for a restricted family of measures, see e.g. [3]-[5],[26].

Figure 1: The explicit open question in arXiv:0910.5850, page 9.

The old proof needs Δ_2 in an integration-by-parts estimate. We replace that step with the bounded-overlap sparse estimate of Lešnik, Roskovec, and Soudský [2]. Their Corollary 1.3 gives constants $C_n, K_n > 0$ such that for every $u \in C_c^2(\mathbb{R}^n)$ there is a positive averaging operator

$$Tz = \sum_{A \in \mathcal{A}} \left(\frac{1}{|A|} \int_A z \right) \mathbf{1}_A, \qquad \sum_{A \in \mathcal{A}} \mathbf{1}_A \leq K_n,$$

for which

$$|\nabla u|^2 \leq C_n\, T(|\nabla^2 u|)\, T(|u|). \tag{3}$$

Concluding, the essence of the method is contained in the following crucial theorem.

Theorem 1.2. *There is positive constant C depending only on dimension n such that for each $u \in \mathcal{C}_c^2(\mathbb{R}^n)$ and each $i \in \{1, \dots, n\}$ there exists a countable family $\mathcal{P}$ of measurable subsets of $\mathbb{R}^n$ satisfying*

$$\left|\frac{\partial u}{\partial x_i}\right|^2 \leq C \sum_{P \in \mathcal{P}} \left(\fint_P \left|\frac{\partial^2 u}{\partial x_i^2}(t)\right| dt \fint_P |u(t)| dt \right) \chi_P \tag{8}$$

and

$$\sum_{P \in \mathcal{P}} \chi_P \leq 5\chi_{\mathbb{R}^n}.$$

In particular, from Theorem 1.2 it follows.

Corollary 1.3. *There are two positive constants C and K depending only on dimension n such that for each $u \in \mathcal{C}_c^2(\mathbb{R}^n)$ there exists a countable family $\mathcal{P}$ of measurable subsets of $\mathbb{R}^n$ such that*

$$|\nabla u|^2 \leq C T_{\mathcal{P}} |\nabla^2 u| T_{\mathcal{P}} |u|$$

and

$$\sum_{P \in \mathcal{P}} \chi_P \leq K \chi_{\mathbb{R}^n}.$$

Figure 2: The sparse pointwise estimate and bounded-overlap property in arXiv:2403.07096, page 4.

2 The sparse-transfer theorem

A *Young function* below is finite, convex, nondecreasing, vanishes at zero, and is not identically zero. No doubling assumption is made.

Theorem 1 (Lebesgue case)**.** *Let $\Omega \subset \mathbb{R}^n$ be open. Let M be an N-function, not necessarily in Δ_2, and let P, Q be Young functions satisfying (1). There is a constant*

$$B_n = 2C_n K_n^2,$$

depending only on the dimension and the constants in (3), such that every $u \in C_c^2(\Omega)$ and every $\theta > 0$ satisfy

$$\int_\Omega M(|\nabla u|)\, dx \leq \int_\Omega P(\theta |\nabla^2 u|)\, dx + \int_\Omega Q(B_n |u|/\theta)\, dx. \tag{4}$$

If P and Q are N-functions, then

$$\|\nabla u\|_{L^M(\Omega)} \leq 4\sqrt{B_n}, \sqrt{\|\nabla^2 u\|_{L^P(\Omega)} \|u\|_{L^Q(\Omega)}}. \tag{5}$$

Proof. Extend u by zero to $\mathbb{R}^n$ and take the operator T in (3). Write

$$g = |\nabla u|, \qquad h = |\nabla^2 u|, \qquad f = |u|, \qquad a = \frac{\theta}{K_n}, \quad b = \frac{2C_n K_n}{\theta}.$$

Thus $ab = 2C_n$. At a point where $g > 0$, (3) gives $ab(Th)(Tf) \geq 2g^2$. Apply (1) with $s = g$, $v = aTh$, and $w = bTf$:

$$M(g) \leq \frac{1}{2} \frac{M(g)}{g^2} (aTh)(bTf) \leq \frac{1}{2} \{M(g) + P(aTh) + Q(bTf)\}.$$

After absorbing $M(g)/2$, this becomes

$$M(g) \leq P(aTh) + Q(bTf). \tag{6}$$

The claim is automatic where $g = 0$.

For every Young function R and nonnegative z, bounded overlap and two applications of Jensen's inequality give

$$\int_{\mathbb{R}^n} R(Tz/K_n)\,dx \le \int_{\mathbb{R}^n} R(z)\,dx. \tag{7}$$

Indeed, at each point the at most K_n averaging terms, supplemented by zeros, form a convex combination; integration then counts every set at most K_n times. Apply (7) first to $R = P, z = \theta h$ and then to $R = Q, z = B_n f/\theta$. Since

$$aTh = T(\theta h)/K_n, \qquad bTf = T(B_n f/\theta)/K_n,$$

integrating (6) proves (4).

For the norm assertion, put $H = \|\nabla^2 u\|_{L^P}$ and $U = \|u\|_{L^Q}$. Assuming $H, U > 0$, apply (4) to

$$\widetilde{u} = \frac{u}{\theta H + B_n U/\theta}.$$

The defining modular property of the Luxemburg norm and monotonicity give $\rho_M(\nabla\widetilde{u}) \le 2$. Convexity of M then gives $\rho_M(\nabla\widetilde{u}/2) \le 1$, hence

$$\|\nabla u\|_{L^M} \le 2\left(\theta H + \frac{B_n}{\theta}U\right).$$

Minimizing over $\theta > 0$ proves (5). The cases $H = 0$ or $U = 0$ follow by the usual limiting argument (and give $\nabla u = 0$). □

The proof isolates the exact weighted input it needs.

Theorem 2 (Modular sparse transfer)**.** *Let $d\mu$ be a measure on $\mathbb{R}^n$. Suppose that every sparse operator T arising in (3) satisfies, for fixed $S_P, S_Q < \infty$,*

$$\int P(Tz/S_P)\,d\mu \le \int P(z)\,d\mu, \qquad \int Q(Tz/S_Q)\,d\mu \le \int Q(z)\,d\mu \tag{8}$$

for all nonnegative z. Then (2) holds with $L = 1$ and $B = 2C_n S_P S_Q$, without any Δ_2 assumption on M. If P, Q are N-functions, the corresponding norm estimate holds with constant $4\sqrt{2C_n S_P S_Q}$.

Proof. Repeat the preceding proof with $a = \theta/S_P$ and $b = 2C_n S_P/\theta$. Their product is $2C_n$; (6) is unchanged, and (8) sends the two terms to $P(\theta h)$ and $Q(2C_n S_P S_Q f/\theta)$. □

Corollary 3 (Globally comparable densities)**.** *Let $\Omega \subset \mathbb{R}^n$ be open and $d\mu = w(x)dx$, where $0 < m \le w \le W < \infty$ almost everywhere. Put $\kappa = W/m$. Under the hypotheses of Theorem 1, both (2) and its norm consequence hold without Δ_2, with*

$$L = 1, \qquad B = 2C_n K_n^2 \kappa^2.$$

Proof. Extend w outside Ω while retaining the same bounds. For every Young function R, convexity, (7), and $\kappa = W/m$ give

$$\int R\left(\frac{Tz}{\kappa K_n}\right) w, dx \le W \int R\left(\frac{T(z/\kappa)}{K_n}\right) dx \le \kappa \int R(z/\kappa) w, dx \le \int R(z) w, dx.$$

Thus Theorem 2 applies with $S_P = S_Q = \kappa K_n$. □

3 Scope and remaining obstruction

Theorem 1 covers every N-function triple used for the source's norm conclusion and keeps its general Young relation (1); it is not restricted to $P = Q = M$ or to inverse-function identities. For the source's modular-only conclusion, however, it adds convexity of P, Q, whereas the source permits merely measurable nondecreasing functions.

More importantly, the Hardy hypothesis in [1] controls the single multiplier $|\nabla\varphi|$ in the P-modular. It does not visibly imply the two uniform sparse modular estimates (8), one of which is needed in the Q-modular. The paper's power and power–exponential examples can have unbounded density oscillation, so Corollary 3 does not cover them. Eight focused upgrade attempts included truncation of M, high-elasticity counterexample families, weighted sparse averaging, and a Hardy-to-sparse implication. No proof or counterexample for arbitrary nondoubling Hardy weights was obtained.

A bounded literature search through 11 August 2026 found no statement of the three-function absorption theorem above. The sparse estimate and its one-function modular contraction are from [2]; their combination with (1), including the exact absorption and weighted transfer criterion, is the new step claimed here.

References

[1] A. Kałamajska and K. Pietruska-Pałuba, *On a variant of Gagliardo–Nirenberg inequality deduced from Hardy*, Bull. Pol. Acad. Sci. Math. 59 (2011), 133–149; arXiv:0910.5850.

[2] K. Lešnik, T. Roskovec, and F. Soudský, *Gagliardo–Nirenberg inequality via a new pointwise estimate*, arXiv:2403.07096 (2024), especially Theorem 1.2, Corollary 1.3, and the modular observation in Section 4.